

Nafiz Imtiaz Khan

Davis, California, United States

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Summary

PhD candidate in Computer Science at UC Davis, advised by Professor Vladimir Filkov. Research focuses on open-source software sustainability, health informatics, and applied data science. Specializes in fine-tuning deep neural networks and large language models, building agentic retrieval-augmented generation (RAG) systems, and developing task-specific AI/ML pipelines.

Education

University of California, Davis

PhD in Computer Science | MSc in Computer Science (2025)

Master's degree conferred July 2025, CGPA: 3.91/4.00

September 2023 – Present

California, United States

Military Institute of Science and Technology (MIST)

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

Technical Skills

Programming Languages: Python, SQL, C/C++, Java, JavaScript, PL/SQL, Typescript, HTML / CSS, Bash

Frameworks/Libraries : PyTorch, Keras, Tensorflow, Hugging Face Transformers, LangChain, Scikit-learn, Pandas, NumPy, Matplotlib, BeautifulSoup, Pyvis, Node, React

ML/AI Techniques: Transformer architectures, fine-tuning, PEFT, LoRA/QLoRA, multimodal & vision-language models (VLMs), retrieval-augmented generation (RAG), benchmarking & evaluation

Tools/Platform: Ollama, Flask, Git, AWS Bedrock, Google Colab, GCP, Docker, Kubernetes, Grafana, Selenium, Tableau, Microsoft Azure, JIRA

Database: ChromaDB, FAISS, MongoDB, FireBase, MySQL, Oracle, SQLite, Neo4j

Professional Experience

Artificial Intelligence Intern

BetterHelp

June 2026 – Present

San Jose, California, United States (Remote)

- Contributing to AI/ML initiatives as part of the Artificial Intelligence Intern team at BetterHelp.

Human Frontier Collective Specialist – GenAI (Internship)

Scale AI, Inc.

August 2025 – September 2025

Remote – California, United States

- Designed and critiqued complex, domain-specific problem sets to rigorously test cutting-edge generative AI models, identifying limitations and refining performance.
- Collaborated with Scale researchers in interdisciplinary sessions to analyze model behavior, shape AI research directions, and explore experimental applications.

Graduate Student Researcher

University of California, Davis

September 2023 – Present

California, United States

- Built **AngioVision** (<https://github.com/Nafiz43/AngioVision>), a multimodal **large language model (LLM)** pipeline for angiography report generation and validation; fine-tuned a **vision-language** framework integrating a **CLIP** alignment module, embedding, and decoder layers, achieving a **10% F1 gain** over the prior SOTA, **Claude 4.6**.
- Led the end-to-end development of **RepoWise** (<https://repowise.github.io/RepoWise-website/>), a domain-aware **retrieval-augmented generation (RAG)** system for OSS governance analytics; benchmarked and demonstrated 84% accuracy on factual queries, outperforming general-purpose LLMs (**GPT-4o**, **Claude 4.5 Sonnet**) on both factual and non-factual repository-analysis tasks.
- Spearheaded the development of **OSSPREY** (<https://oss-prey.github.io/OSSPREY-Website/>), a tool to assess open-source project health via real-time analytics with an average latency of 3.5s and 124MB memory usage per project, integrating **Vue.js** (frontend), **TensorFlow** (deep learning), and **LangChain & Ollama** (LLM inference pipeline design)
- Engineered **ReACT-GPT** (<https://github.com/Nafiz43/ReACT-GPT>), an **LLM-and-RAG**-powered framework for extracting actionable insights from scientific articles, achieving a 75% BERTScore by integrating **LangChain** (orchestration), **ChromaDB** (vector retrieval), and **Ollama** (lightweight LLM inference)
- Built **EvidenceBot** (<https://nafiz43.github.io/EvidenceBot/>), a privacy-preserving **RAG** tool for querying and extracting insights from a diverse set of documents, using **Streamlit** and the **ReACT-GPT** framework
- Created **PCL-Fetcher** (<https://nafiz43.github.io/PCL-Fetcher/>), an LLM-powered pipeline automating procedural case log documentation from radiology reports, achieving an 86% F1 score—exceeding the crosswalk benchmark by over 13%
- Designed **MammoGen-RAG** (<https://github.com/Nafiz43/MammoGen-RAG>), a multimodal RAG pipeline leveraging **vision-language models (VLMs)** to automate mammography report generation, achieving a 90% F1 score through integrated visual and textual reasoning
- Improved foundational language model performance in **LLM-as-Developers** (<https://github.com/zeerakb1/llm-developers-evaluation>) by 10% (BERTScore) on answering StackOverflow questions via **parameter-efficient fine-tuning (PEFT)** and **QLoRA**
- Developed and optimized **Transformer-** and **LSTM-based** models for forecasting Open Source Software (OSS) project sustainability (<https://github.com/OSS-PREY/OSSPREY-Pex-Forecaster>), achieving an 8% improvement in F1 score over SOTA

- Built the **AFMC Admission Test** Module, a full-stack software solution for the AFMC Entrance exam, utilizing **Node.js**, **Bootstrap**, and **MongoDB** to generate unique randomized questions for 30,000 candidates and automate the evaluation and ranking process
- Spearheaded a team to develop **C-Archive** using **PHP**, **MySQL**, **SQL**, and **Bootstrap** to store ongoing events of the department [[GITHUB](#)]

Patents

1. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Systems And Methods For Artificial Intelligence-Driven Evidence-Based Action" (USPTO Filing ID: 32417/70825P) [[REF](#)]

Award & Honors

- **Best Paper Award** for "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," by Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, and Vladimir Filkov, at the 2nd International Conference on Software Engineering of Emerging Technologies (**SEET 2026**)
- **Champion** in the **Medical Robotics Challenge for Contagious Diseases**, organized by **Imperial College London** [[REF](#)]
- **Champion** in the Creative App Contest of the **Tri Robo Cup Championship** organized by the **MIST Robotics club** [[REF](#)]
- **3rd runner-up** at **COVID-19 Combatants Unification Competition**, organized by **IEEE Bangladesh Chapter** [[REF](#)]
- **Top Downloaded Article Award** - in the year 2021, and 2022; provided by **Engineering Reports Journal - Wiley**
- **GGCS Summer Ph.D Fellowship** - 2024, awarded by Graduate Group of Computer Science, UC Davis
- Recognized for academic excellence at MIST, achieving placement on the **Dean's List of Honor** and receiving the **Merit Scholarship** in 2018, 2019, and 2020

Publications

[\[Google Scholar\]](#)

1. Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, Vladimir Filkov, "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," Software Engineering of Emerging Technologies, Communications in Computer and Information Science (CCIS), Springer, pp. 268-288, 2026. (**Best Paper Award**) [[LINK](#)]
2. **Nafiz Imtiaz Khan**, Kiley Cleland, Vladimir Filkov, and Roger Eric Goldman, "Intelligent Documentation in Medical Education: Can AI Replace Manual Case Logging?," JAMIA Open, 2026. [[LINK](#)]
3. Arjun Ashok*, **Nafiz Imtiaz Khan***, Swati Singhvi, Stefan Stanciulescu, Zhouhao Wang, Vladimir Filkov (*equal contribution), "Multi-Ecosystem Modeling of OSS Project Sustainability," Journal of Systems and Software (JSS), 2026. [[LINK](#)]
4. **Nafiz Imtiaz Khan**, Priyal Soni, Arjun Ashok, Vladimir Filkov, "OSSPREY: AI-Driven Forecasting and Intervention for OSS Project Sustainability," ASE 2025 Tool Demonstrations (IEEE/ACM International Conference on Automated Software Engineering), 2025. [[LINK](#)]
5. **Nafiz Imtiaz Khan**, Vladimir Filkov, "EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions," FSE 2025 Tool Demonstrations (Foundations of Software Engineering), 2025. [[LINK](#)]
6. Zeerak Babar, **Nafiz Imtiaz Khan**, Muhammad Hassanain, Vladimir Filkov, "Open-Source LLMs for Technical Q&A: Lessons from StackExchange," International Conference on Software Engineering of Emerging Technologies (SEET), Springer, pp. 615-626, 2025. [[LINK](#)]
7. **Nafiz Imtiaz Khan**, Vladimir Filkov, Roger Eric Goldman, "AI-Driven Automation of Procedural Case Log Documentation," AMIA 2025 Clinical Informatics Conference (abstract), 2025. [[LINK](#)]
8. **Nafiz Imtiaz Khan**, Vladimir Filkov, "From Models to Practice: Enhancing OSS Project Sustainability with Evidence-Based Advice," FSE 2024 (Foundations of Software Engineering), 2024. [[LINK](#)]
9. Amrin Rahman, Mahedi Kamal, Afrina Zinia, Amrin Akhter Pinky, Sabrina Afrin Toma, Abedur Rahman Khan, M Akhtaruzzaman, **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, "AI Integrated Entertaining and Monitoring System for Daily Care of Elders," Proceedings of the 3rd International Conference on Computing Advancements, 2024. [[LINK](#)]
10. Md. Saddam Hossain Mukta, Jubaer Ahmed, Mohaimenul Azam Khan Raiaan, Nur Mohammad Fahad, Muhammad Nazrul Islam, Nafiz Imtiaz, Md. Adnanul Islam, Mohammed Eunus Ali, Sami Azam, "Behavior Based Group Recommendation from Social Media Dataset by Using Deep Learning and Topic Modeling," SN Computer Science, 2024. [[LINK](#)]
11. Muhammad Nazrul Islam, Raiyan Jahangir, Nasif Shahriar Mohim, Md. Wasif-Ul-Islam, Anika Ashraf, **Nafiz Imtiaz Khan**, Mohammad Ratul Mahjabin, Abu Saleh Musa Miah, Jungpil Shin, "A Multilingual Handwriting Learning System for Visually Impaired People," IEEE Access, 2024. [[LINK](#)]

12. Nazia Shehnaz Joynab, Muhammad Nazrul Islam, Ramiza Rumaisa Aliya, A.S.M. Rakibul Hasan, **Nafiz Imtiaz Khan**, Iqbal H. Sarker, "A federated learning aided system for classifying cervical cancer using PAP-SMEAR images," Informatics in Medicine Unlocked, 2024. [\[LINK\]](#)
13. Muhtasim Fuad, B M Nishanul Alam Nishat, Tashfia Fatema, Mohian Islam, Faria Habib, Fahim Foysal, Saba Ahmed, **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, "A Blockchain-based Framework to Develop a Sustainable Supply Chain Management System for Apparel Industry," 2023 IEEE 9th International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), 2023. [\[LINK\]](#)
14. Raiyan Jahangir, Nasif Shahriar Mohim, **Nafiz Imtiaz Khan**, Md Akhtaruzzaman, Muhammad Nazrul Islam, "Proposing Novel Recurrent Neural Network Architectures for Infant Cry Detection in Domestic Context," 2023 IEEE 11th Region 10 Humanitarian Technology Conference (R10-HTC), 2023. [\[LINK\]](#)
15. Muhammad Nazrul Islam, **Nafiz Imtiaz Khan**, Toki Tahmid Inan, Iqbal H. Sarker, "Designing User Interfaces for Illiterate and Semi-Literate Users: A Systematic Review and Future Research Agenda," Sage Open, 2023. [\[LINK\]](#)
16. **Nafiz Imtiaz Khan**, Tahasin Mahmud, and Muhammad Nazrul Islam, "COVID-19 and Black Fungus: Analysis of Public Perceptions through Machine Learning," Engineering Reports, 2022. (IF = 2 and SJR Rank = Q2) [\[LINK\]](#)
17. Akib Zaman, Mohammad Shahjahan Majib, Shoeb Ahmed Tanjim, Shah Md. Ahasan Siddique, Shafayetul Islam, Md Shadman Aadeeb, **Nafiz Imtiaz Khan**, Riasat Haque, Md Rashid Ul Islam, M. Rayhan Ferdous Faisal, Siddharth Malik, and Muhammad Nazrul Islam, "UVC-PURGE: A Novel Cost-Effective Disinfection Robot for Combating COVID-19 Pandemic," IEEE Access, 2022. [\[LINK\]](#)
18. Samia Zakir Sarothi, Khondaker Sakil Ahmed, **Nafiz Imtiaz Khan**, Aziz Ahmed, and Moncef Nehdi, "Predicting Bearing Capacity of Double Shear Bolted Connections Using Machine Learning," Engineering Structures, 2022. (IF = 5.582 and SJR Rank = Q1) [\[LINK\]](#)
19. Samia Zakir Sarothi, Khondaker Sakil Ahmed, **Nafiz Imtiaz Khan**, Aziz Ahmed, and Moncef Nehdi, "Machine Learning-Based Failure Mode Identification of Double Shear Bolted Connections in Structural Steel," Engineering Failure Analysis, 2022. [\[LINK\]](#)
20. Muhammad Nazrul Islam, Sumaiya Nuha Mustafina, Tahasin Mahmud, and **Nafiz Imtiaz Khan**, "Machine learning to predict pregnancy outcomes: a systematic review, synthesizing framework and future research agenda. BMC pregnancy and childbirth," BMC Pregnancy and Childbirth, 2022. (IF = 3.09 and SJR Rank = Q1) [\[LINK\]](#)
21. Md Mahbubar Rahman, **Nafiz Imtiaz Khan**, Iqbal H. Sarker, Mohiuddin Ahmed, and Muhammad Nazrul Islam, "Leveraging machine learning to analyze sentiment from COVID-19 tweets: A global perspective," Engineering Reports, 2022. [\[LINK\]](#)
22. Faria Habib, Tasfia Fatema, Munswarim Khan, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "Exploring Design Attributes and Development of an Acoustic VR Game to Improve Ethical Values of Visually Impaired People," 2022 IEEE 7th International Conference for Convergence in Technology (I2CT 2022), 2022. [\[LINK\]](#)
23. Moumita Bhowmik, Naim Ibna Khadem Al Bhuyain, Md. Rokonzaman Reza, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "Neurophysiological Feature Based Stress Classification Using Unsupervised Machine Learning Technique," International Conference on Fourth Industrial Revolution and Beyond (IC4IR 2021), 2022. [\[LINK\]](#)
24. Muhammad Nazrul Islam, **Nafiz Imtiaz Khan**, Nafiz Islam, Samuli Laato and A. K. M. Nazmul Islam, "Monitoring the Health and Movement of Quarantined COVID-19 Patients with Wearable Devices," Healthcare Technology Solutions for Pandemics – A Roadmap (Springer Book Chapter), 2022. [\[LINK\]](#)
25. Raiyan Jahangir, Md. Wasif-Ul-Islam, Nasif Shahriar Mohim, Anika Ashraf, **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, "Towards Developing a Voice-Over-Guided System for Visually Impaired People to Learn Writing the Alphabets," 2022 25th International Conference on Computer and Information Technology (ICCIT), 2022. [\[LINK\]](#)
26. Muhammad Nazrul Islam, Tahasin Mahmud, **Nafiz Imtiaz Khan**, Sumaiya Nuha Mustafina, A. K. M. Najmul Islam, "Exploring machine learning algorithms to find the best features for predicting modes of childbirth," IEEE Access, 2021. [\[LINK\]](#)
27. Mohammad Shahjahan Majib, Md. Mahbubur Rahman, T. M. Shahriar Sazzad, **Nafiz Imtiaz Khan**, Samrat Kumar Dey, "VGG-SCNet: A VGG Net-Based Deep Learning Framework for Brain Tumor Detection on MRI Images," IEEE Access, 2021. (IF = 3.367 and SJR Rank = Q1) [\[LINK\]](#)
28. Jesika Rahmana, Khondaker Sakil Ahmed, **Nafiz Imtiaz Khan**, Kamrul Islam, Sujith Mangalathu, "Data-driven shear strength prediction of steel fiber reinforced concrete beams using machine learning approach," Engineering Structures, 2021. (IF = 5.582 and SJR Rank = Q1) [\[LINK\]](#)
29. Anika Islam Aishwarja, Nusrat Jahan Eva, Shakira Mushtary, Zarin Tasnim, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "Exploring the Machine Learning Algorithms to Find the Best Features for Predicting the Breast Cancer and its Recurrence," 3rd International Conference on Intelligent Computing and Optimization (ICO 2021), 2021. [\[LINK\]](#)

30. Deepon Deb Nath, **Nafiz Imtiaz Khan**, Jesmin Akhter and Abu Sayed Md. Mostafizur Rahaman, "Prediction of Android Malicious Software using boosting Algorithms," 4th International Conference on Emerging Technologies in Computing (iCETiC 2021), 2021. [[LINK](#)]
31. Noor Nafiz Islam, **Nafiz Imtiaz Khan**, Md Abdur Razzak, and Muhammad Nazrul Islam, "Design, Development and Evaluation of a Physical Exercise Monitoring and Managing System for Athletes," 23rd International Conference on Information Integration and Web Intelligence (iiWAS 2021), 2021. [[LINK](#)]
32. Ifath Ara, Tasneem Mubashshira, Fariha Fardina Amin, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "Towards Developing a Mobile Application for Detecting Intoxicated People through Interactive UIs," International Joint Conference on Advances in Computational Intelligence (IJCAI 2021), 2021. [[LINK](#)]
33. Muhammad Nazrul Islam, **Nafiz Imtiaz Khan**, Ayon Roy, Md Mahbubar Rahman, Md Saddam Hossain Mukta, A.K.M Najmul Islam, "Sentiment Analysis of Bangladesh-specific COVID-19 Tweets using Deep Neural Network," 62nd International Conference on Information Technology and Management Science (ITMS 2021), 2021. [[LINK](#)]
34. Md. Rokonzaman Reza, Fabiha Mukarrama Binte Mannan, Dhrubo Barua, Shafayetul Islam, **Nafiz Imtiaz Khan**, and Sharifa Rania Mahmud, "Developing a Machine Learning Based Support System for Mitigating the Suppression Against Women and Children," 5th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT 2021), 2021. [[LINK](#)]
35. Shutonu Mitra, Tasfia Tasnim, Arr Rafi Islam, **Nafiz Imtiaz Khan**, and Shajahan Majib, "A Framework to Detect and Prevent Cyberbullying from Social Media by Exploring Machine Learning Algorithms," 6th International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2 2021), 2021. [[LINK](#)]
36. Muhammad Rafsun Sheikh, Tarek Hasan Masud, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "An Efficient Transfer Learning Model for Predicting Forged (Handwritten) Signature," 2021 International Conference on Computer, Communication, Chemical, Materials, and Electronic Engineering (IC4ME2), 2021. [[LINK](#)]
37. **Nafiz Imtiaz Khan**, Tahasin Mahmud, Muhammad Nazrul Islam, and Sumaiya Nuha Mustafina, "Prediction of Cesarean Childbirth using Ensemble Machine Learning Methods," 22nd International Conference on Information Integration and Web-based Applications & Services (iiWAS 2020), 2020. [[LINK](#)]
38. Tarannum Zaki, **Nafiz Imtiaz Khan**, and Muhammad Nazrul Islam, "Evaluation of User's Emotional Experience through Neurological and Physiological Measures in Playing Serious Games," 20th International Conference on Intelligent Systems Design and Applications (ISDA 2020), 2020. [[LINK](#)]
39. **Nafiz Imtiaz Khan**, Sumaiya Nuha Mustafina, Farzana Faruk Jhumu, A.H.M Zobyer, Masrur Hasan Mahin, Md. Ariful Islam Tarek, Raiyan Rahman, and Muhammad Nazrul Islam, "Towards Developing an Automated Attendance Management System using Fingerprint Sensor," International Conference on Emerging Technology in Computing, Communication and Electronics (ETCCE 2020), 2020. [[LINK](#)]
40. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Leveraging Language Models to Discover Evidence-Based Actions for OSS Sustainability," (Under review). [[LINK](#)]
41. Raiyan Jahangir, **Nafiz Imtiaz Khan**, Amritanand Sudheerkumar, Vladimir Filkov, "MammoWise: Multi-Model Local RAG Pipeline for Mammography Report Generation," (Under review). [[LINK](#)]
42. Lin Zhou, **Nafiz Imtiaz Khan**, Charlie Schweik, Vladimir Filkov, "Open-Source Software Sustainability: Defining and Measuring a Multifaceted Concept," (Under review). [[LINK](#)]